

MARANDA 2025 KCSE PREDICTION



EXPECTED EXAM 1 POWER MECHANICS

447/2

PAPER 2 (PRACTICAL)

TIME: 2½ HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Kenya Certificate of Secondary Education.

Instructions to candidates

- (a) Write your name and index number in the spaces provided above.
- (b) Sign and write the date of examination in the spaces provided above.
- (c) There are **TEN** stations in this examination.
- (d) Candidates are allowed **15 minutes** at each station
- (e) At each station, candidates are not allowed to either review the previous stations' work or read instruction for other stations
- (f) Attempt **ALL** exercises in each station
- (g) All dimensions are in millimeters unless otherwise stated.

For examiner's use only

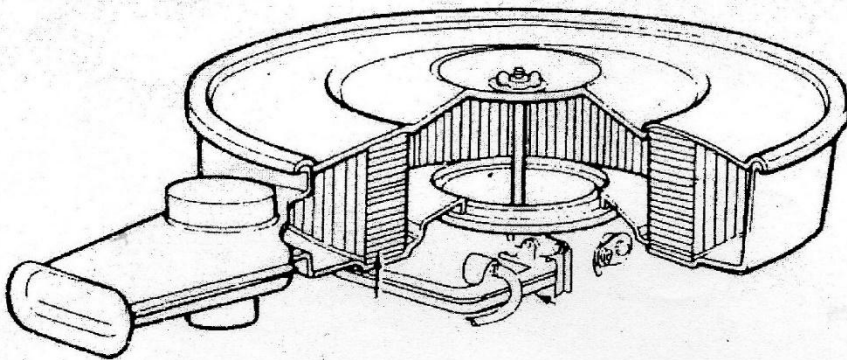
Stations	1	2	3	4	5	6	7	8	9	10	Total
Marks											

STATION 1

INSTRUCTIONS:

The figure below shows a truncated air cleaner assembly. On the drawing paper (page 3) provided, sketch in good proportion the exploded view of the assembly and label **four** parts.

(10 marks) a well-drawn exploded view with the following components: air cleaner cap, air intake tube, stud, air cleaner gasket, vacuum motor, carburetor air cleaner, air cleaner element temperature sensor kit



STATION 2

INSTRUCTIONS:

Draw an exploded view of the differential unit and label all the parts.

(10 marks)

A well-drawn exploded view with the following parts:

-Shaft, axles ,side bevel gear, crown wheel, differential cage, and the pinion gear---10 marks

STATION 3

INSTRUCTIONS:

On the single cylinder engine provided;

(a) Remove the exhaust valve and show it to the examiner

(b) Reassemble the valve to its original position

(c) Measure and record the exhaust valve clearance.

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(d) Name the tool you have used to take the reading

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(10 marks)

Let the examiner check your work

The examiner to check the candidate work and award marks accordingly.

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STATION 4

- (b) Identify the fasteners labelled P to T and state a feature that makes each fastener unique in its class. (5 marks)

FASTENER	NAME	UNIQUE FEATURE
P		
Q		
R		
S		
T		

(10 marks)

STATION 5

INSTRUCTIONS:

Table 3

Cell Number	Specific Gravity	Battery Voltage
1		
2		
3		
4		
5		
6		

Identify two defects on the fan belt and state one possible cause of each defect; (10 marks)

(a) Measure and record the specific gravity and battery voltage of each cell in Table

3. (Take cell number 1 to be next to the positive terminal)

Table 3 (b) Check and comment on the state of:

(i) Battery electrolyte level

(ii) Battery charge

The examiner to check and award marks accordingly

STATION 6

INSTRUCTIONS:

On the engine provided, carry out the following tasks. (10 marks)

- a) Remove the rocker cover.
- b) Adjust the valve clearance for number 1 to given specifications.
- c) Replace the rocker cover.

Let the examiner check your work.

Mark and award marks accordingly.

STATION 7

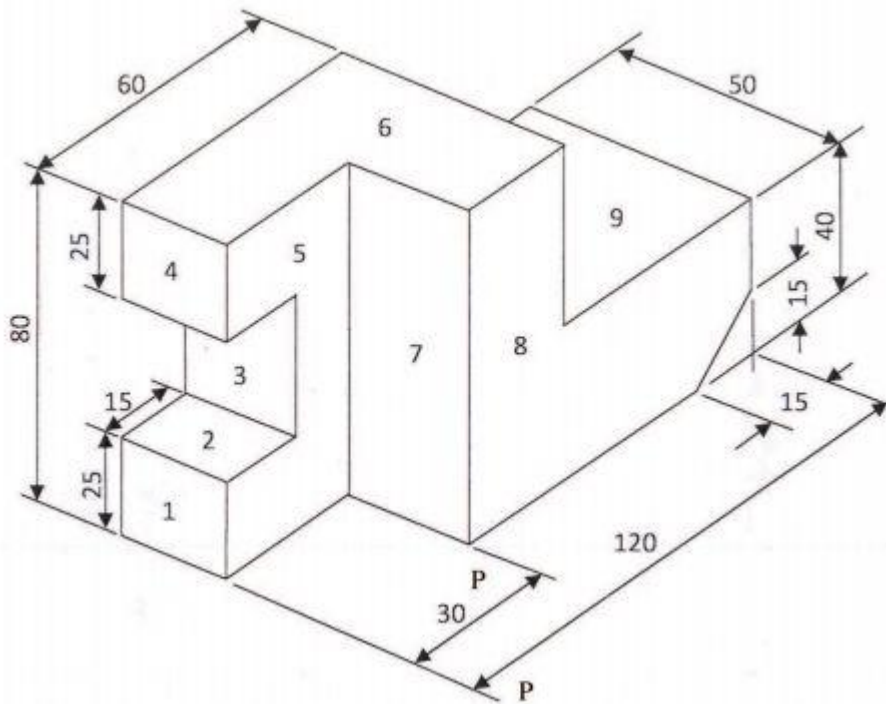
INSTRUCTIONS:

COMPONENT	DEFECT	EFFECT
V — Engine mount	Broken	Excessive vibration; Judder
W — Fuel filter	Clogged	Insufficient fuel delivery
X — Cylinder head gasket	Burnt or worn	No compression
Y — Distributor cap	Cracked	Engine won't start
Z — Inlet valve	Worn/bent	No compression
5 x O = 20 marks	5 x D = 20 marks	5 x 1 = 5 marks

STATION 8

INSTRUCTIONS:

The figure 1 below shows 3 orthographic views of a block drawn in the first angle projection. Draw the block FULL SIZE in isometric projection taking P as the lowest point. (10 marks)



STATION 9

INSTRUCTIONS:

Using the instruments provided measure and record the readings in the table below.

PART	DESCRIPTION OF MEASUREMENT	MEASUREMENT TAKEN	NAME OF TOOL USED	ACCURACY OF TOOL USED
A	Cam height			
B	Depth of blind hole			
C	Intake valve margin			
D	Flapper			
E	Pitch of spark plug			

STATION 10

INSTRUCTIONS: using the tools and equipment provided make the scoop below.

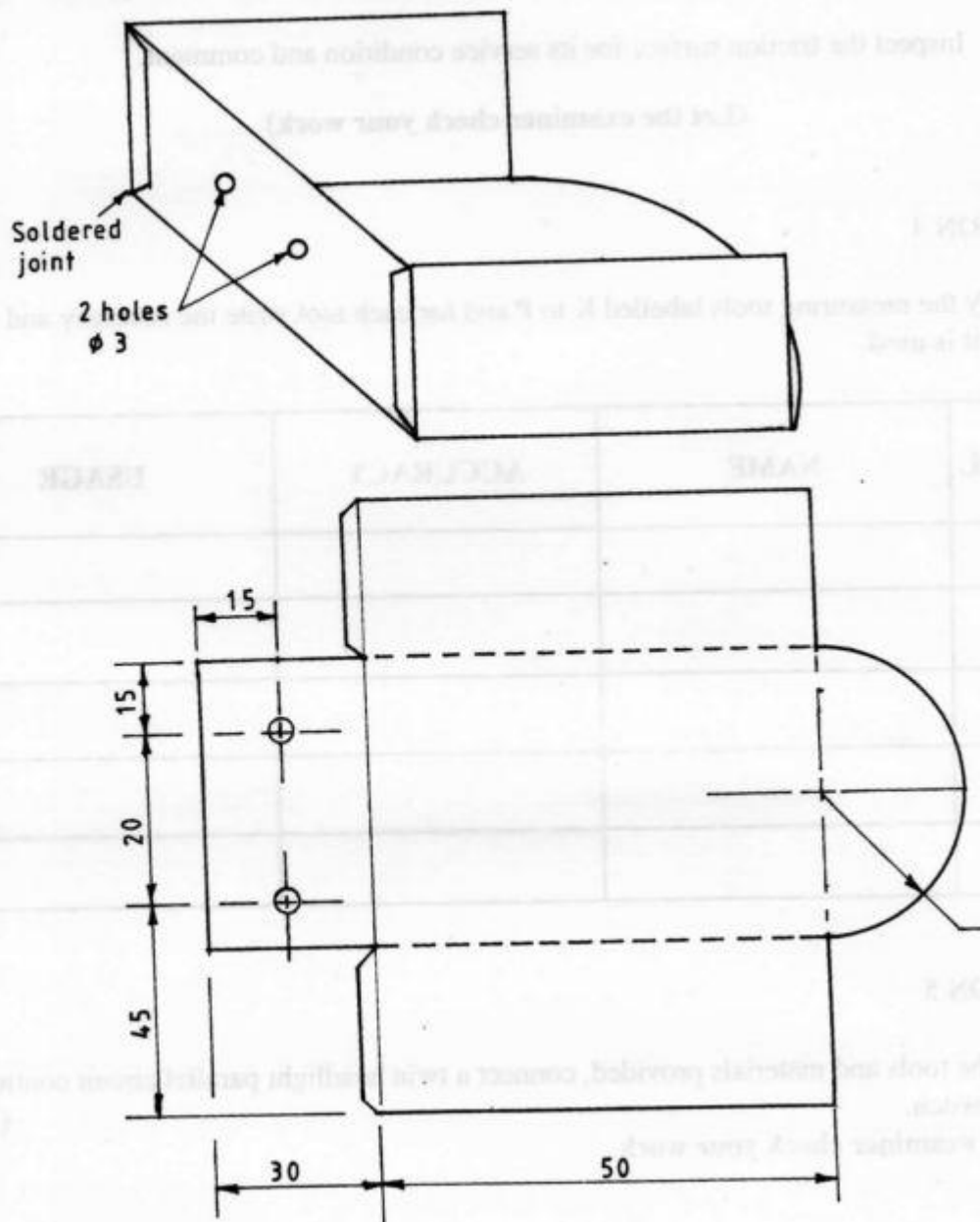


Figure 2

Award marks accordingly.